NYAH BENTLEY 2026

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Normal LCD (SLM On)

The TIE reconstructions were next performed with the SLM actively modulated using Arduino-generated patterns. Figures 8- 25 present TIE reconstructions obtained under patterned illumination for both intact and damaged LCD conditions. Despite differences in the specific patterns applied, all reconstructions exhibit consistent behavior: the imposed intensity patterns introduce structured variations in the recovered phase and surface height maps. In the case of the intact LCD (Figures 8-10), periodic features corresponding to the projected patterns are clearly visible, even in regions where no physical surface variation is present. This indicates that the TIE algorithm is interpreting intensity modulation as phase structure, resulting in artificial surface features.

For the damaged LCD (Figures 11-25), these effects are further amplified. Existing defects interact with the imposed patterns, producing exaggerated peaks and valleys in the reconstructed surface. As a result, defects appear significantly more pronounced compared to the unmodulated case, demonstrating that structured illumination enhances both real and artificial features.

Additionally, variations in pattern spatial frequency influence the appearance of the reconstruction. Higher-frequency patterns produce more localized and rapidly varying features, while lower-frequency patterns result in smoother, wave-like structures. These results confirm that the observed depth variations are not solely due to physical surface geometry but are strongly influenced by intensity gradients introduced by the display.

Normal LCD (SLM On)

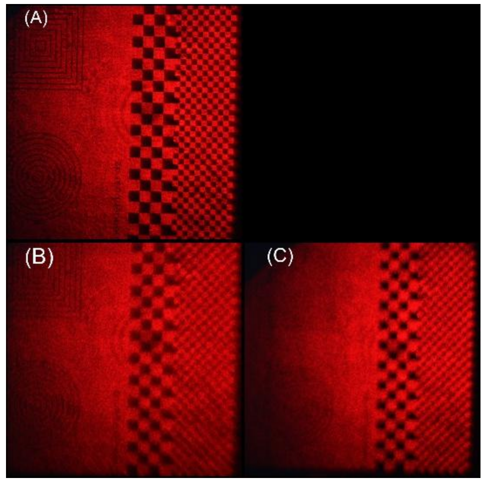

Figure 8 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying Squares on the right-side pattern on a damaged screen. (A) In-focus intensity image I_0. (B) Intensity image acquired 5 mm ahead of the focal plane ($+z$). (C) Intensity image acquired 5 mm behind the focal plane ($-z$).

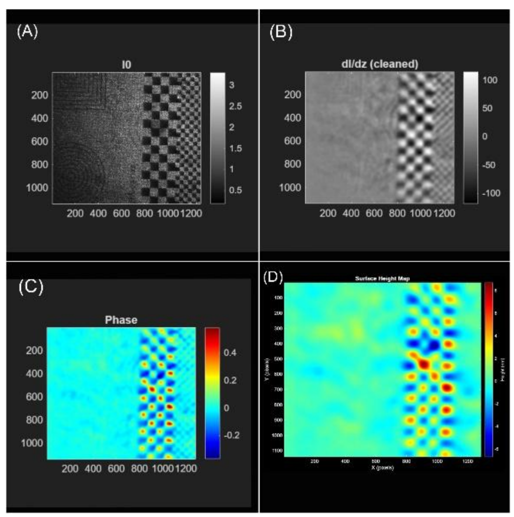

Figure 9 (A) shows the in-focus intensity I_0, serving as the reference image with surface texture and illumination variations. (B) shows the axial intensity derivative $\partial I/\partial z$, highlighting edges and structural features. (C) shows the recovered phase distribution, highlighting optical path variations in the LCD. Planar tilt was removed to isolate structural features, with minor noise present in low-intensity regions due to small I_0values. (D) presents the corresponding two-dimensional height map.

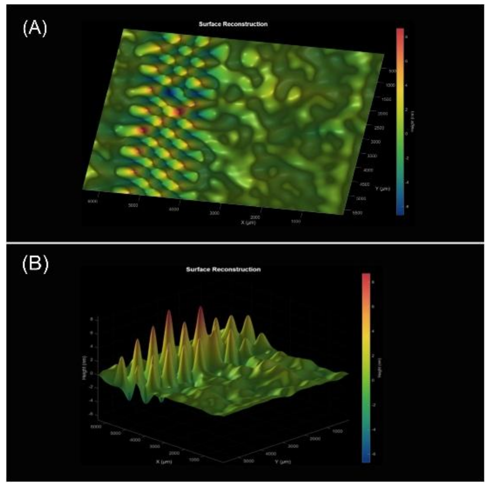

Figure 10 displays the three-dimensional surface reconstruction of a normal LCD with the SLM displaying a pattern, shown (B) side and (A) top views. Peaks and valleys correspond to texture features in the original intensity images.